Trade Credit Guarantee Networks

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Keywords: Social networks, Trade credit guarantee

JEL classification numbers: D 85, G 29, O17.

First Draft: Please do not cite, or quote.

30 July 2015

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Abstract

We develop a model of strategic trade credit guarantee networks that focuses on favour exchanges, where producers act as guarantors for trade credit to one another. Our analysis highlights the architecture of informal trade credit guarantee networks when the cost of forming a link is positive and benefits are frictionless. We find that only cycle-free tree networks are pair wise stable and are efficient. Further pair wise stable sub networks may co-exist even in the absence of enforcement problems and the number of components in the network is an increasing step function of the cost of forming a link.

1. Introduction.

In a world where actors are not anonymous, economic and social networks often play a major role in shaping economic outcomes[2]. Consider, as an example, an industry where firms face idiosyncratic demand shocks. In any given period, some firms may face a high demand in their output market, while others face a low demand. To the extent that firms depend not only on their own capital, but also on credit to finance production, it is quite possible that in the absence of a smoothly functioning credit market, firms who face a high demand scenario may not be able to take full advantage of this high demand. In such cases, the existence of social and economic networks that facilitate transfer of this excess credit available to a producer who faces a low demand state to one who faces a high demand state is clearly welfare enhancing. In this paper, we study such networks.

To fix ideas, consider the following example drawn from our field studies in the informal sector garment manufacturing industry in the Metiabruz area of Kolkata, a city in the eastern part of India. The garment manufacturing industry in Metiabruz, Kolkata is possibly the largest non-branded readymade garment industry in India. It supplies manufactured garments to wholesalers across the country. Garment manufacturers, who are locally known as *Ostagars,* usually buy

[2] There is a vast literature on the economics of networks that focuses on how social and economic networks affect individual incentives and, in turn, shapes economic outcomes. See, for example, for the effect of social and economic networks on labour market outcomes (Granovettor (1973), Montgomery (1991), Bian (1997), Calvó-Armengol (2000), Yakubovich (2005), X. Zuo *et al* (2014)); for consumers' purchasing decisions for buying a particular product (Ellison and Fudenberg (1995), Kranton and Minehart (2000, 2001)); for R & D decisions taken by firms (Goyal and Moraga Gonzalez, 2001); for market competition (Goyal and Joshi , 2003). See also the very insightful survey articles by Jackson (2014) and Goyal (2015).

their inputs, mainly fabric, thread, and other variable inputs from a small set of local large merchants who are locally known as *Mahajans*. Provision of trade credit in the input market is a common feature among the sellers of inputs: buyers of inputs buy part of their total purchases, or the entire amount, on credit from the sellers against promises to repay by some pre-specified date. These supplier credit transactions are not backed by formal, legally enforceable contracts; everything is informal, and the transacting parties simply keep notes on the amounts and dates. However, suppliers of credit appear, in general, to recover the dues from their debtors without any great difficulty. The main point is that the system works fairly smoothly outside the formal legal system[3].

Trade credit is not unique in this industry. There is ample international evidence of suppliers who extend trade credit to their buyers. Fisman and Raturi (2004) study trade credit extension by firms in Sub-Saharan African countries and find that competitive firms are more likely to provide credit to their buyers than monopolistic firms. This is a "locking in" device that firms use to prevent their customers from moving away to other suppliers, though trade credit is offered after the usual waiting period when creditworthiness is determined. McMillian and Woodruff (1999) found exactly the opposite result in Vietnam where firms that face lower competition are more likely to offer trade credit and these trade credit relationships are based on family ties or friendship ties[4]. Petersen and Rajan (1997) found that in the United States, trade credit is offered to small firms with low access to credit from formal credit institutions. Fafchamps (1997) reports

[3] This feature of this industry is perhaps the primary reason that makes detailed, careful studies of the operation of the markets that co-exist in this industry so difficult, as outsiders investigators are viewed with a great deal of suspicion. For example, we found it virtually impossible even to draw up a list of final producers to draw random samples, leave alone find hard data on which individuals acted as guarantors for whom. Our example, therefore, is based on extraction of what appears to be common knowledge in the industry through repeated free-form interviews of participants in the industry.

[4] McMillian and Woodruff report that in Vietnam, smaller firms face difficulties in obtaining credit from the formal lenders and, hence, rely greatly on trade credits.

that all types of firms in Zimbabwe depend on trade credit, though there may be some bias against providing such credit to African owned firms[5]. Fafchamps (2000) reports that in Kenya and Zimbabwe, where some ethnic groups play a dominant role in manufacturing, blacks and women 'are disadvantaged in the attribution of supplier credit'. Fishman (2003) finds that among African firms, a firm is more likely to get supplier credit from a firm owned by someone with the same ethnic origin than from a firm owned by someone with a different ethnic origin.

In Metiabruz, input suppliers provide trade credit to those input buyers with whom they have some form of, often long term, connections. Unlike in the African countries, neighborhood and ethnic ties do not appear to determine access to trade credit. Input sellers are typically Hindu or Jain *Marwaris*, and the manufacturers are typically Bengali Muslims who live in geographically separated neighborhoods and need not be friends outside their professional domain. Sometimes, suppliers provide trade credit to producers who do not belong to their direct networks if a guarantor guarantee the repayment of the loan. Usually producers who take trade credit from a particular input seller, and have close ties with the latter, are the potential guarantors for producers who do not have close links with the input seller.

A producer offers to act as a guarantor only if he thinks that the borrower who wants the guarantee is a creditworthy, and is likely to repay his debt to the input seller as he is responsible for repayment in the event of a default. The amount of loan that can be guaranteed by a guarantor depends on the guarantors' own credit surplus, which is the difference between his credible loan limit and the amount of trade credit that he has actually taken from his lender. Lenders forms estimates of the credible limits for loans to their borrowers based on the past business

[5] See also Cheng and Pike (2003), Fafchamps and Lund (2003), Fisman and Love (2003), Burkart, and Ellingsen (2004), Atanasova (2007), Fafchamps and Gubert (2007), Cull *et al* (2009), Gianetti, Burkart, and Ellingsen (2011).

transactions between them. In a similar manner, guarantors form estimates about how much trade credit can be guaranteed based on their private information about the borrowers' business.

In this paper, we construct a formal model of trade credit guarantee networks, i.e., networks of producers who can act as guarantors for each other[6]. This extends the usual trade credit network literature by incorporating a complementary network of potential recipients of trade credit which effectively increases the trade credit available to individual producers in the high demand states. Since networks play such an important role, the central focus of this paper is to model the architecture of strategically stable trade credit guarantee networks observed in the input markets of this industry. In particular, we model the incentives to form such networks when forming links is costly and requires mutual consent. We explore the architecture of stable networks and their efficiency properties. Our model is shorn of any form of extraneous ethnic, friendship or kinship ties as we do not explore the formation of these networks, the features common to the people in a particular network, who is included and who is left out, or indeed even centrality or leadership roles. In spite of this, we find that (pair wise) stable networks can exist, that the industry may be partitioned into disjoint (pair wise) stable sub-networks, each with its own particular architecture, and that there is no tension between stability and efficiency, a problem that plagues much of the literature.[7]

Our paper is a contribution to the theory of strategic network formation. It also contributes to the literature on how the presence of social and economic networks shape the economic outcomes, by providing one example, which shows, how in the presence of idiosyncratic demand shocks,

[6] The formal model is an undirected network of producers: links between two producers require the consent and effort of both parties.
[7] See, for example, Dutta and Muthuswami (1997), Jackson (2005, 2008), Bloch and Dutta (2008).

the trade credit guarantee networks observed in the readymade garment industry in Metiabruz, Kolkata affect the provision of informal trade credit extended by input market participants among themselves. In that sense, our paper is related to the extensive literature on mutual insurance networks (see, for example, Coate and Ravallion (1993), Murgai, Winters, Sadolet and de Janvry (2002), Fafchamps and Lund (2003), Genicot and Ray (2003), De Weerdt and Dercon (2006), Fafchamps and Gubert (2007), Bramoulle and Kranton (2007), Bloch, Genicot and Ray (2008)).[8] The underlying presumption, based on empirical observation, is that such networks are often based on kinship ties or geographical proximity. In the more recent theoretical literature, (Bramoulle and Kranton (2007), Bloch, Genicot and Ray (2008)) insurance ties are bilateral, though a person is free to have as many such ties as he pleases, and the reciprocity arrangements are self enforcing. In our model, kinship or other kinds of ties that bring to bear external pressures to form ties, induce adherence to promised commitments play no role: people work in the same industry, perhaps work in a narrow geographical proximity, know each other, share information and invest in maintaining professional relationships. To keep the model simple, we assume that enforcement of arrangements, i.e., offering to underwrite trade credit and repayments of such debt is not a problem, and postpone a more elaborate model building in the enforcement issues in a repeated game structure for future work[9] . [10]

A third important feature of our model is that benefits from network membership do not decay with the number of links separating two individuals as in Jackson and Wolinsky (1996), Bala and Goyal (2000), Jackson and Watts (2002), Watts (2001), or as an extreme example, in Bramoulle

[8] See Genicot and Ray (2003), footnote 3 for an extensive discussion of the related empirical literature.
[9] See Belleflamme and Bloch (2004) for a similar assumption.
[10] See Jackson, Rodriguez-Barraquer and Xu Tan (2012) for a gift exchange model where gift exchange norms are internally enforceable.

and Kranton (2007), Bloch, Genicot and Ray (2008) where only players who are directly linked can help each other. In this paper, we assume that the role of the network is to allow friends to introduce friends to other friends, who in turn introduce the first person to more friends, and so on till a guarantor can be found[11]. That is, a producer act as a guarantor for another one if they are path connected in a network: a potential guarantor does not discriminate between his direct neighbours and indirect neighbours.

Even so, we find that the network comprising of all players may be partitioned into disjoint components (more on this later) where a variety of architectural structures can co-exist simultaneously. Further, the size of such sub-networks, measured by the number of players in each one, falls as the cost of forming links rises.

The rest of the paper is organized as follows. The basic features of the trade credit guarantee network is presented in the section 2. In section 3 we collect some related formal definitions from the existing literature, which we use in this model. In section 4 we present the network formation and architecture of strategically stable networks and their efficiency properties. Section 5 concludes.

2. Basic features.

Let there be $N = \{1,2,....,n\}$, $n > 1$, *ex ante* identical producers in the market. Producers sell their output in identical independent markets, where demand can be low or high with probabilities $p > 0$ and $(1-p) > 0$, respectively. We assume that these demand shocks are uncorrelated across the output markets.

[11]See Bala and Goyal (2000), Galeotti and Goyal (2002), Galeotti, Goyal, and Kampherst (2006) for similar mechanism where benefits are frictionless, i.e., benefits do not decay with number of links separating any two individuals in a network.

Producers buy input goods from input sellers who supply inputs against cash payments and extend some credit to buyers if required. Producers buy in cash and take credit from the input suppliers only if they run out. For simplicity, we assume that each producer deals regularly with a single input seller, but individual input sellers may deal with more than one producer.

Designate the demand for trade credit, i.e., the credit that a producer wants from his input supplier by $L_l \geq 0$ in the low demand state and by $L_h > 0$ in the high demand state. Let $L_m > 0$ be the trade credit limit that the input supplier offers to a producer, where $L_l < L_m < L_h$.

If a producer faces a high demand shock in his output market, then he would need some additional credit, $F = (L_h - L_m) > 0$, over and above what his own input supplier is willing to provide him. A producer, who faces a low demand shock in the output market, has a surplus of available trade credit $(L_m - L_l)$, over and above his own needs. This he can transfer to other producers if he acts as a guarantor for the latter if they purchase inputs from his own input seller.

Assumption 1 [A1]. The profit function of producer i, $i \in N$, in the high demand state, designated by $\pi_i(L_{cg})$, is upward rising and concave in the size of the trade credit guarantee, L_{cg} with $\pi_i(0) > 0$.

Since, by definition, L_h is the maximum trade credit that the producer demands in the high demand state, $\pi_i(L_{cg})$ attains its maximum at F, hence $0 \leq L_{cg} \leq F$. For computational simplicity, we assume that $F = L_m - L_l$.

3. Some Definitions.

A producer has access to trade credit guarantee from another producer only if he is linked to the latter directly, or via common friends in the same line of work. To formalize this notion, and for analytical purposes, in this section we define the idea of a social network and collect some related definitions that we shall need in one place. The primary reference is Jackson (2008) from which we draw much of what follows in this section.

Definition 1. A pair $(i, j) \in N \times N$ and designated by ij is called a *link* between producers i and j.

Definition 2. A set of pairs $ij \in N \times N$ and designated by $g(N)$ is called a *set of links*, or edges on the set of nodes N.

Definition 3. The pair $(N, g(N))$ is called a *network*, or graph, on N. Where there is no possibility of confusion, we shall suppress N and write this simply as $g(N)$.[12]

In this paper, the set of producers in the industry is the set of nodes[13] N. Let $g(N) - ij$ denote the network obtained by severing an existing link between nodes i and j from network $g(N)$, while $g(N) + ij$ is the network obtained by adding a new link between nodes i and j in network $g(N)$.

Networks may be classified as directed networks and undirected networks. In undirected networks, both parties in any link consent to the link formation, e.g., friendships. In directed networks, mutual consent is not needed, e.g., citations of professional papers, legal judgments, etc.

[12] See Diestel (2005) for a more elaborate exposition of graph theory.
[13] In this paper the terms node, player and producer are synonymous.

Definition 4 (Jackson 2008). An *undirected network* $(N, g(N))$ is a pair of sets $N = \{1,2,3,........,n\}$ and $g = \{ij\}_{\substack{i \in N \\ j \in N \\ i \neq j}}$ with $ij = ji$ for all $i, j \in N$ and $i \neq j$. It is said to be *directed* if $ij \neq ji$ for all $i, j \in N$ and $i \neq j$.

Since, a trade credit guarantee can flow in either direction in a link, depending upon which producer needs the trade credit guarantee and which producer is able to provide it, our focus in this paper is on undirected networks among producers. Henceforth, in this paper by the term network we mean an undirected network. These networks provide a foundation for our analysis of trade credit guarantee flows from one producer to another in the sense described above. Two producers i and j, $i \neq j$ in N may not be directly linked in a network. However, they may still be linked via friends, friends of friends, and so on. The next two definitions formalize the idea of such indirect links.

Definition 5 (Jackson 2008). A *walk* in a network $(N, g(N))$ between nodes i and j is a sequence of links $i_1 i_2, i_2 i_3, ..., i_{K-1} i_K$ such that $i_k i_{k+1} \in g(N)$ for all $k \in \{1,2,3......., K-1\}$ with $i_1 = i$ and $i_k = j$.

Definition 6 (Jackson 2008). A *path* in a network $(N, g(N))$ between nodes i and j is a sequence of links $i_1 i_2, i_2 i_3, ..., i_{K-1} i_K$ such that $i_k i_{k+1} \in g(N)$ for all $k \in \{1,2,3......., K-1\}$ with $i_1 = i$ and $i_k = j$, such that each node in the sequence is distinct. In this case, we say producers i and j are path connected.

Notice that a path is a special case of a walk.

Assumption 2 [A2]: If a path in the network $(N, g(N))$ exists between producers i and j then the two producers can serve as guarantors for each other.

Definition 7. A *sub network* $(M, g(M))$ of the network $(N, g(N))$ is a sub graph restricted to the set of nodes $M \subset N$ so that $ij \in g(M)$ where $i \in M$, $j \in M$ and $i \neq j$.

It says that the sub network $(M, g(M))$ is obtained by deleting all links except those that are between nodes in M.

Assumption 3[A3]: Producers in any path connected sub network $(M, g(M))$ of the network $(N, g(N))$ who are in a position to act as guarantors do not discriminate between their immediate neighbours and other producers in M when providing the guarantees to those who want them.

What this assumption says is that if two players are path connected then the number of links separating them has no impact on the provision of trade credit guarantee provided by one to the other. In other words, trade credit guarantee connections extend beyond direct neighbours to friends of friends without any loss of strength. We do not consider the case where the amount guaranteed, or the probability that a guarantee will be available, decays as the number of links separating two producers increases.

Definition 8. A *component* of a network $(N, g(N))$ is a non-empty sub network $(M, g(M))$ such that $\phi \neq M \subseteq N$, $g(M) \subseteq g(N)$, where (1) $(M, g(M))$ is path connected, and (2) if $[i \in M$ and $ij \in g(N)]$ then $[j \in M$ and $ij \in g(M)]$.

Let $i \in N$. For any $j \in N$ with $i \neq j$ the cost of forming the link ij is $c \geq 0$. We designate a sub network of the network $(N, g(N))$ containing the node i by $(M_i, g(M_i))$. Designate the number of nodes in M_i by $m \geq 1$. Suppose that producer i faces a high demand in its output market. Let the number of producers in M_i, *other than producer i*, who face high demand in their respective output markets be n_h and number producers who face low demand in their respective output markets be n_l, where $n_h + n_l = m - 1$. Designate the gross expected profit of producer i by $E\pi_i(g(M_i))$.

A social network may change overtime if producers have incentives to add or remove links. The following definition gives us conditions under which such changes will not occur in a network. In the literature a network is said to remain unaltered if it is stable. The next definition provides one commonly used definition of network stability.

Definition 9. Consider the path connected sub network $(M_i, g(M_i))$, any pair $i \neq j \in M_i$, cost to a player of forming a link, $c > 0$, and gross expected profits $E\pi_i(g(M_i))$ and $E\pi_j(g(M_i))$, respectively, where $|M_i| = m$. The path connected sub network is said to be *pair-wise stable* if the following hold:

i) If $ij \in g(M_i)$, then $E\pi_i(g(M_i)) - E\pi_i(g(M_i) - ij) \geq c$, and
$E\pi_j(g(M_i)) - E\pi_j(g(M_i) - ji) > c$

and ii) If $ij \notin g(M_i)$, and $E\pi_i(g(M_i) + ij) - E\pi_i(g(M_i)) > c$, then
$E\pi_j(g(M_i) + ji) - E\pi_j(g(M_i)) < c.$[14]

[14] The underlying assumption here is that a producer can delete an existing link unilaterally. What this definition does not consider is that producers may add or delete more than one link at the same time as in a link announcement game.

Definition 10 (Jackson 2008). A walk $i_1i_2, i_2i_3, i_3i_4, ..., i_{K-1}i_K$ in a network $(N, g(N))$ is called a *cycle* when it starts and ends at the same node, i.e., $i_1 = i_K$, and such that all other nodes are distinct ($i_k \neq i_{k'}$) where $k < k'$ unless $k = 1$ and $k' = K$.

Definition 11 (Jackson 2008). A *tree* is a path connected sub network that has no cycles.

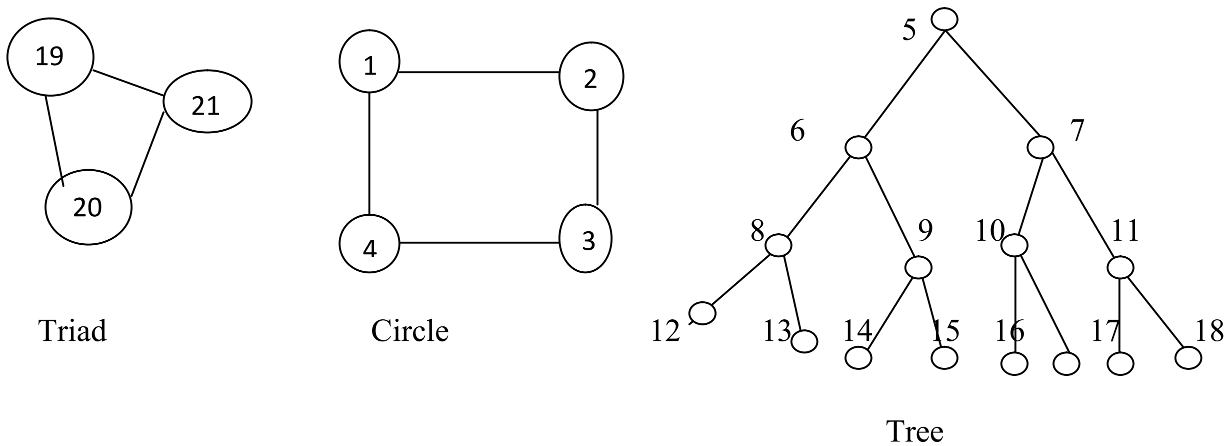


Figure 1. Triad, Circle, and Tree

A network may be stable but not *efficient* in the sense that another network over the same set of producers may provide higher total benefit to the members. Definition 11 formalizes the notion of efficiency that we shall use.

Definition 12. Consider any two networks g and $\bar{g}$ on the set of players N, where $g \neq \bar{g}$. Let $W(g)$ be the sum of gross expected individual profits generated by the network g. A network g is called *efficient* if and only if $W(g) - 2c\eta_g \geq W(g) - 2c\eta_{\bar{g}}$, where η_g and $\eta_{\bar{g}}$ are total number of bilateral links in network g, and $\overline{g}$, respectively, and $2c$ is the total cost of forming a bilateral link.[15]

4. Network Structure.

In this section we return to our trade credit guarantee model and explore the architecture of the trade credit guarantee networks**.**

4.1. An example.

To see what assumptions $A2$ and A3 mean, we begin with an example where the benefit declines as the number of links between two players increases. One can think of this as the benefit that one can expect from a relationship, dampened by a declining probability of receiving the benefit as the distance between the players increases. We show that as the rate of decay changes, the architecture of stable network changes and at the low level of decay producers have less incentive to form links with those producers with whom they are already path connected. Benefits are displayed next to each node in the diagram.

Example 1. Let $N = \{1,2,3,4\}$ and suppose that all players are path connected. The gross expected profit to a producer (before the cost of link formation is deducted) from a direct link is 100, and this decays by the factor $\delta < 1$ for each *indirect* link in the shortest path connecting a producer to another one. Thus, if along the shortest path two producers have k links separating them, where $k = 1,2,3$, the gross benefit to each one from the connection is: $\sum_{k=1}^{3} \delta^{k-1} 100$. Let $c = 25$.

[15] An alternative definition of efficiency in Social Network Theory is Pareto Efficiency. See Jackson (2008) for details.

We consider four cases. In case A (Figure 2), $\delta = 0.5$; in case B (Figure 3), $\delta = 0.8$; in case C (Figure 4), $\delta = 0.9$, and in case D (Figure 5), $\delta = 1$.

Case A: $\delta = 0.5$

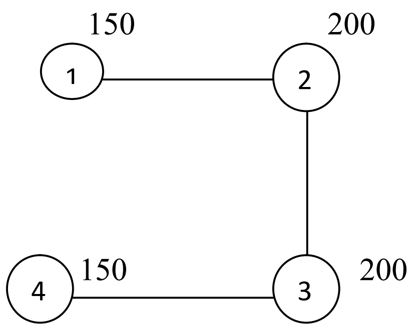


Figure 2: (i)

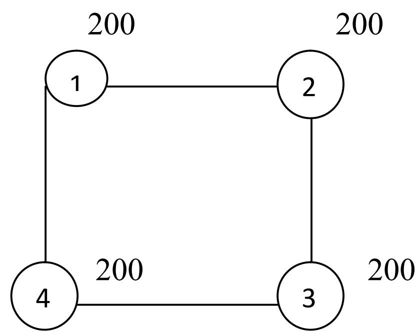


Figure: 2 (ii)

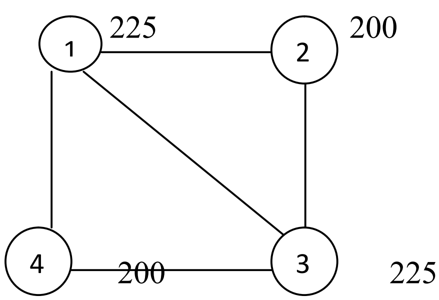


Figure 2: (iii)

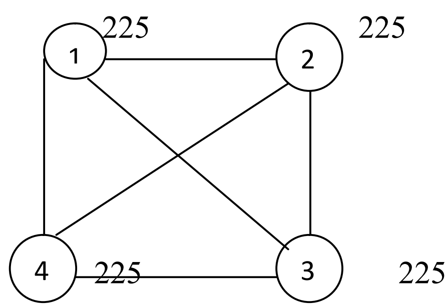


Figure: 2 (iv) pair wise stable

Figure 2. Case A: Pair wise stability when $\delta = 0.5$

Consider Figure 2(i). Producer 1 has one direct link with producer 2, and two indirect links, one with producer 3, and the other with producer 4. His gross benefit from being a member of the network equals $100 + 50 + 25 = 175$. Deducting the cost of his single direct link, his net benefit from being a member of this network is $175 - 25 = 150$. However, producer 2 has two direct links and one indirect link, so his net benefit from being a member of the network is $100 + 100 + 50 - 50 = 200$. A similar calculation yields a net benefit of 200 for producer 3 and of 150 for producer 4. Using a similar argument we can compute the net benefit to each producer when the number of direct links changes. The payoff that a player gets is displayed next to his node in Figure 2.

Notice first that the architecture of the network in panel 2(i) is not pair wise stable as an additional link formation between producers 1 and 4 increases each one's net benefit to 200 without affecting the net benefits of the other two producers. However, inspection of the remaining panels in Figure 2 shows that the only pair wise stable architecture under this parameter configuration is the complete network (Figure 2(iv)).

Case B: where $\delta = 0.8$

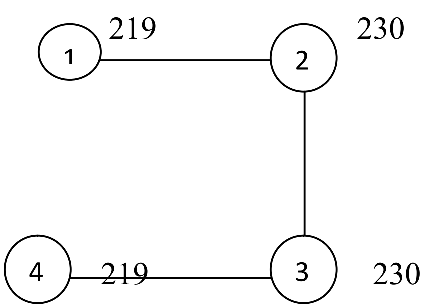


Figure 3: (i)

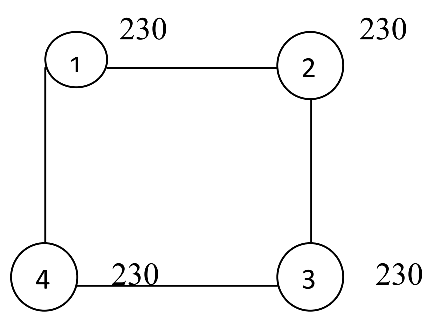


Figure: 3 (ii) pair wise stable

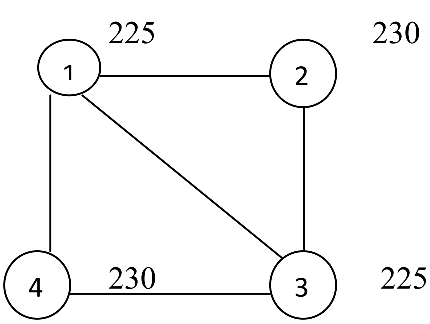


Figure 3: (iii)

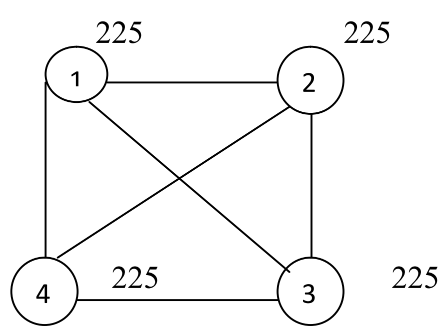


Figure: 3 (iv)

Figure 3. Case B: Pair wise stability when $\delta = 0.8$

Consider now case B, when $\delta = 0.8$. As before, we show the relevant network architectures in Figure 3, with the net benefits for each producer marked next to the relevant node. A simple inspection of the panels shows that the cycle network in panel 3(ii) is the only pair wise stable network.

Case C: $\delta = 0.9$

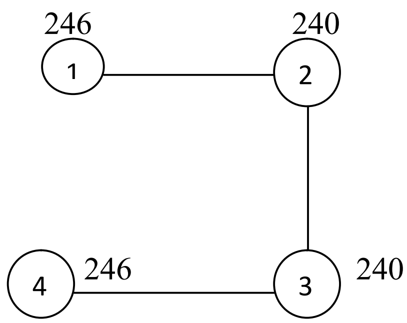


Figure 4: (i) pair wise stable

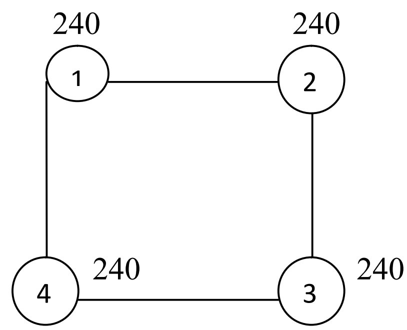


Figure: 4 (ii)

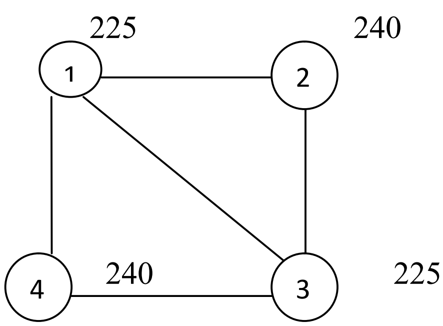


Figure 4: (iii)

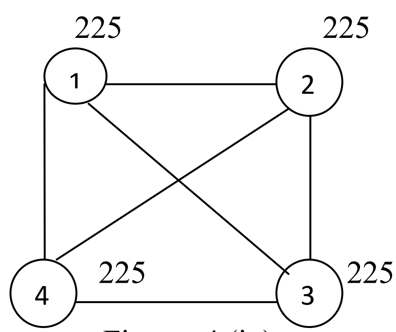


Figure: 4 (iv)

Figure 4. Case C: Pair wise stability when $\delta = 0.9$

In case C, when $\delta = 0.9$, the only pair wise stable network is the line network (Figure 4(i)).

Finally, we look at the case D, where $\delta = 1$.

Case D: $\delta = 1$

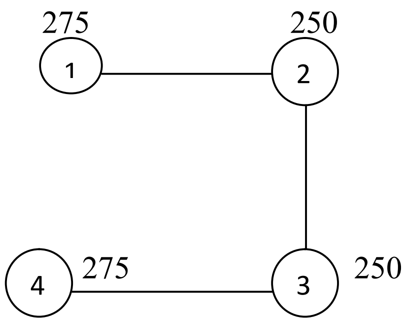


Figure 5: (i) pair wise stable

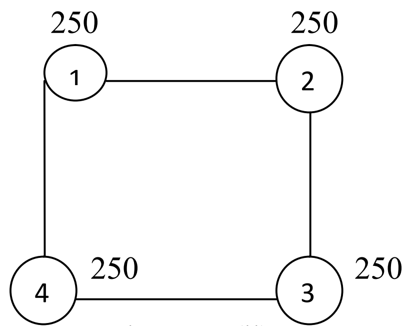


Figure: 5 (ii)

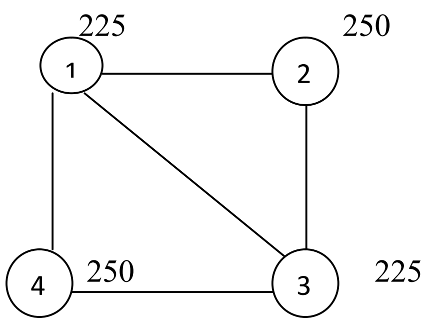


Figure 5: (iii)

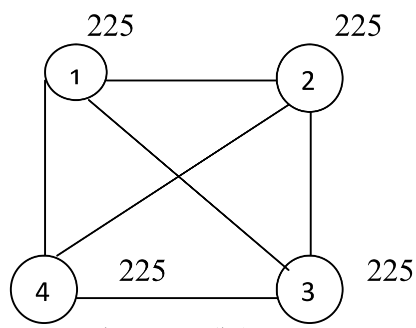


Figure: 5 (iv)

Figure 5. Case D: Pair wise stability when $\delta = 1$

In case D, when $\delta = 1$, the only pair wise stable network is the line network (Figure 5 (i)).

What this example shows is that given the cost of forming an additional link, as the value of δ increases the net expected profit from forming additional links declines, so that the stable network architecture changes in the sense that producers form fewer direct links. Further as the value of δ approaches 1 the architecture of stable network stabilizes to a tree network. The general point is that assuming that $\delta = 1$, as in our model, yields a stable network architecture that could be robust for small perturbations in the value of δ from unity and, hence, our assumption need not be seen to be particularly stringent[16].

In addition to the cost of forming links, the architecture of a network depends on how the benefit that is generated by the network is distributed among the network members. In this paper, producers in the industry form expectation about their profit from being a member of a network based on how much trade credit guarantee they require in each state of demand, and how much they can expect to get in a good state given the requirements of other players and the availability of trade credit guarantees.

4.2. Trade Credit Guarantee.

Suppose that there is only one producer in M_i, say producer i, who gets a high demand shock. As no other producer in M_i other than producer i faces a high demand state, producer i gets the full amount of loan guarantee that he wants. More generally, as long as $n_h - 1 \leq n_l$, producer i gets the full amount of loan guarantee that he wants. Formally, the maximum trade credit guarantee that producer i will get is given by

$$\hat{F} = \min[F, F\frac{n_l}{n_h}] \text{ if } n_h > 0$$

$$= F \text{ otherwise} \tag{1}$$

Since, n_h and n_l are not known *a priori* to producer i, he does not know how much trade credit guarantee he can get from the network and, hence, forms expectation regarding how much profit

[16] At the other extreme is the assumption that benefits can only be exchanged between direct neighbours: here $\delta = 0$. See Bloch, Genicot and Ray (2008) for such an assumption.

he can earn by being a member of the network in the following way. In a low demand state in his output market, we designate his profit by π_l and in a high demand state by $\pi(L_{cg})$, when he receives the amount $L_{cg}, F \geq L_{cg} \geq 0$, of trade credit guarantee. We assume that $\pi(0) > \pi_l$. Suppose that producer i is an isolated producer. Here, $M_i = \{i\}$. Since there is only one player in the component hence the number of potential guarantors for i is zero. His expected profit from being a member of the empty sub-network $(M_i, g(M_i))$ is then given by: $E\pi_i(g(M_i)) = p\pi_l + (1-p)\pi(0)$ where $p \in (0,1)$ is the probability that he faces a low demand state in the output market[17]. Since, $\pi(0) > \pi_l > 0$, $E\pi_i(g(M_i)) > 0$.

Let $c > 0$ be the cost to each producer of forming a link. We designate the net expected profit of producer i, i.e., his expected profit minus his cost of forming links by $NE\pi_i(g(M_i))$. Suppose now that producer i forms a link with another isolated producer j in $(N, g(N))$. If producer i faces a low demand state in his output market then he does not need a trade credit guarantee from producer j and his profit is π_l. The probability of this happening is p. If, however, producer i gets a high demand shock in the output market, and producer j a low demand shock in his output market, then producer j can provide a trade credit guarantee up to the amount that producer i requires, i.e., F. Producer i's profit then is $\pi(F) > \pi(0)$. The probability of this happening is $(1-p)p$. Therefore, producer i's net expected profit, i.e., his gross expected profit minus the cost of forming the link is given by: $NE\pi_i(g(M_i)) = p\pi_l + (1-p)p\pi(F) + (1-p)^2\pi(0) - c$

(2)

[17]As we assume that the social norm of offering to guarantee trade credit and timely repayment is internally enforceable, looking at one period expected benefit suffices, since along the equilibrium path of the repeated game the same benefit function gets repeated ad infinitum. See Bendor and Mookherjee (1990) for a similar argument.

Since producer j is identical with producer i, producer j has the same net expected profit.

From Assumptions A 2 and A3, we know that as long as two producers are path connected one can act as a trade credit guarantor for the other, i.e., even if two producers are not directly connected being a member of the component $(M_i, g(M_i))$ generates a frictionless[18] network externality. Consider the following example.

Example 2. Suppose that M_i has three producers, who are all path connected. Suppose that producer i has only one direct link in $g(M_i)$. Then the net expected profit of producer i is given by: $NE\pi_i(g(M_i)) = p\pi_l + (1-p)p^2\pi(F) + 2(1-p)^2 p\pi(\frac{F}{2}) + (1-p)^3\pi(0) - c$ (3)

Now suppose that producer i has two direct links in the network $g(M_i)$. Then

$$NE\pi_i(g(M_i)) = p\pi_l + (1-p)p^2\pi(F) + 2(1-p)^2 p\pi(\frac{F}{2}) + (1-p)^3\pi(0) - 2c \qquad (4)$$

where $\pi(\frac{F}{2})$ is the profit of producer i when he gets half of the trade credit guarantee that he requires. Notice that the net expected profit is greater for those with only direct link. Using equations (2) and (4), it is easy to check that player i will from a second link if and only if $p(1-p)^2[\pi(\frac{F}{2}) - \frac{1}{2}\pi(F) - \frac{1}{2}\pi(0)] > \frac{1}{2}c$ [19] . What the example indicates is that there can be upper bounds on the number of direct links that a producer will form within a component given the cost of forming a link.

---

[18] See Bala and Goyal (2000) for the distinction between frictionless network externality and network externality with friction.

[19] By Jensen's Inequality, and Assumption 1, the L.H.S is positive.

Consider the path connected sub network $(M, g(M))$ of the network $(N, g(N))$. The gross expected profit of producer i from the sub network, before deducting his link formation costs, is:

$$E\pi_i(g(M)) = p\pi_l + \sum_{k=1}^{m-1} {}^{m-1}c_k \ p^k (1-p)^{m-k} \pi\{\min(F, \frac{k}{m-k} F)\} + (1-p)^m \pi(0) \tag{5}$$

where $(m-1)$ is the maximum number of potential guarantors that producer i has in this sub network; $k \leq m-1$ is the number of producers who get low demand shocks in their respective output markets among $(m-1)$ producers (excluding producer i); and $(m-k)$ is the number of producers who face high demand shocks in their respective output markets(including producer i).

The first term in equation (5) is the case where producer i faces a low demand shock in the output market. The probability of this happening is p. In this case, producer i does not require trade credit guarantee and receives profit π_l. The second term of equation (5) is the case where some producers $k \leq m-1$(excluding producer i) are in a position to provide trade credit guarantees to other members of the sub network. The joint probability that k producers (excluding producer i) get bad shocks in their respective output markets, and the remaining $(m-k)$ producers (including producer i) get good shocks in their respective output markets, is ${}^{m-1}c_k p^k (1-p)^{m-k}$, where $k = 1,2,...,m-1$. In this case producer i gets a trade credit guarantee equivalent to either F or $\frac{k}{m-k} F$, whichever is smaller. The third term in equation (5) is the case where all m producers face good shocks in their respective output markets. The joint probability of this happening is $(1-p)^m$. Here, producer i gets no trade credit guarantee from the network, and receives a profit $\pi(0)$.

Now m, the number of producers in M, can be even or odd. Simplifying equation (5) we get:

$$E\pi_i(g(M)) = \sum_{k=1}^{\frac{m}{2}-1} {}^{m-1}c_k p^k (1-p)^{m-k} A_k + \frac{1}{m-1} {}^{m-1}c_{\frac{m}{2}} p^{\frac{m}{2}} (1-p)^{\frac{m}{2}} A +$$

$$\sum_{k=\frac{m}{2}+1}^{m-2} (m-1-k) p^{m-k} (1-p)^{m-k} A + p(1-p)A + (1-p)\pi(0) + p\pi_l \qquad (5')$$

when m is an even number, and

$$E\pi_i(g(M)) = \sum_{k=1}^{\frac{m-1}{2}} {}^{m-1}c_k p^k (1-p)^{m-k} A_k + \sum_{k=\frac{m-1}{2}+1}^{m-2} (m-1-k) p^{m-k} (1-p)^{m-k} A + p(1-p)A +$$

$$(1-p)\pi(0) + p\pi_l \qquad (5'')$$

when m is an odd number. (See Appendix for full expressions of equation $(5')$ and $(5'')$.

where $A_k = \pi(\min(1, \frac{k}{m-k})F) - \min(1, \frac{k}{m-k})\pi(F) - \{1 - \min(1, \frac{k}{m-k})\}\pi(0)$; $A = \pi(F) - \pi(0)$.

As we have observed, assuming $\delta = 1$ means that the distance between two producers in a network, measured by the number of links connecting them, has no effect on either the loan guarantee sizes, or on the gross expected profit of a player. The next four theorems establish the properties of the gross expected profit function of an individual producer, say, producer i. Proposition 1 says that given Assumption 1, the expected profit is positive for any sub network size (measured by the number of players in the sub network) containing producer i; Proposition 2 establishes that the marginal gross expected profit is positive if the probability of getting a bad shock is sufficiently high; Proposition 3 demonstrates the existence of network externality; Proposition 4 establishes that the marginal gross expected profit decreases with the number of players in M.

Proposition 1. Assumption 1 implies that $E\pi_i(g(M)) > 0$, $\forall i \in M$ with $m > 1$.

Proof: See appendix.

Suppose now that $\{j\}$ is an isolated node of $(N, g(N))$. Let $M' = M \cup \{j\}$. If a link is formed between some producer in M and producer j, we get the new sub network of $(N, g(N) + ij)$ denoted by, $(M', g(M'))$, where $\left|M'\right| = m + 1$[20]. Suppose m is an even number. Then using equation $(5'')$ the gross expected profit of producer i, $E\pi_i(g(M'))$, from adding this link is:

$$E\pi_i(g(M')) = \sum_{k=1}^{\frac{m}{2}} {}^{m}c_k p^k (1-p)^{m+1-k} A_k' + \sum_{k=\frac{m}{2}+1}^{m} (m+1-k) p^{m-k} (1-p)^{m-k} A +$$

$$(1-p)\pi(0) + p\pi_l \tag{6}$$

where $A_k' = \pi(\min(1, \frac{k}{m+1-k})F) - \min(1, \frac{k}{m+1-k})\pi(F) - \{1 - \min(1, \frac{k}{m+1-k})\}\pi(0).$

See appendix for the full expression of equation (6).

We can find a similar expression for $E\pi_i(g(M'))$ when m is an odd number. This is as follows:

$$E\pi_i(g(M')) = \sum_{k=1}^{\frac{m-1}{2}} {}^{m}c_k p^k (1-p)^{m+1-k} A_k' + \sum_{k=\frac{m-1}{2}+1}^{m} (m+1-k) p^{m-k} (1-p)^{m-k} A +$$

$$(1-p)\pi(0) + p\pi_l \tag{6'}$$

From equations $(5')$ and (6) we get:

$$E\pi_i(g(M')) - E\pi_i(g(M)) = mp(1-p)^m A_1' +$$

[20] Notice that the underlying network on N changes as we add these links with isolated players in (N, g(N)), (N, g(N) + ij), etc.

$$\sum_{k=2}^{\frac{m}{2}} p^{k-1}(1-p)^{m-k+1} \frac{(m-1)(m-2.........(k+1)}{(k-1)(k-2)...3.2.1} (\frac{m}{k} pA_k{}' - A_{k-1}) \tag{7}$$

where $A_1{}' = \pi(\min(1,\frac{1}{m})F) - \min(1,\frac{1}{m})\pi(F) - \{1-\min(1,\frac{1}{m})\}\pi(0).$

See appendix for the derivation of equation (7). Hence, we get the following proposition.

Proposition 2. i) Let m be an even number. Assumption 1 implies that $E\pi_i(g(M')) - E\pi_i(g(M)) > 0$ if $p > \max_k(p_k)$ where $p_k = (\frac{k}{m})(\frac{A_{k-1}}{A_k'})$, and $k = 2,3,4,........,\frac{m}{2}$.

ii) Let m be an odd number. Assumption 1 implies that $E\pi_i(g(M')) - E\pi_i(g(M)) > 0$ if $p > \max_k(p_k)$ where $p_k = (\frac{k}{m})(\frac{A_{k-1}}{A_k'})$, and $k = 2,3,4,.....,\frac{m-1}{2}$.

Proof: See appendix.

What Proposition 2 shows is that the gross expected profit of a producer in the sub network $(M, g(M))$ increases monotonically with m for sufficiently high value of p. Now suppose that producer $i \in M$ does not form the link with producer j, and some producer $s \in M$ forms the link with producer j. The next proposition, Proposition 3, establishes that there is a positive network externality in our model (Jackson (2008), pp-213), i.e., inclusion of an additional, previously isolated, producer belonging to N, in the sub network $(M, g(M))$ increases the expected profit (before deducting the cost of link formation) for all existing producers in the sub network, irrespective of who among them forms the link.

Proposition 3. $E\pi_i(g(M)+sj) > E\pi_i(g(M))$, where $i \in M, s \in M, i \neq j,$ and $\{j\} \cap M = \phi$.

Proof: See appendix.

What Propositions 1, 2, and 3 do not tell us is the nature of the curvature of $E\pi_i(g(M))$ with respect to the numbers of players in the sub network. Consider the following example.

Example 3: Let $N = \{1,2,3,4,5,6\}$. Table 1 provides data on actual profit levels for different values of the credit guarantee received by a producer. The maximum value of $L_{cg} = F = 100$.

Table 1: Profit levels for different values of trade credit guarantee, L_{cg}.

L_{cg}	0	20	25	33.3	50	66.7	100
$\pi(L_{cg})$	600	1860	2200	2500	2980	3400	3460

Let $(N, g(N))$ be an empty network. Consider the path connected sub networks $(M_i^a, g(M_i^a))$, $(M_i^b, g(M_i^b))$, $(M_i^c, g(M_i^c))$, $(M_i^d, g(M_i^d))$, and $(M_i^e, g(M_i^e))$, where $|M_i^0| = 1$; $|M_i^a| = 2$; $|M_i^b| = 3$; $|M_i^c| = 4$; $|M_i^d| = 5$; and $|M_i^e| = 6$[21], containing producer i. Table 2 and Table 3 show the gross expected profits and marginal expected profits, of producer i before deducting his link formation costs for different sub network sizes and for different values of p, respectively.

Table 2: Gross expected profits of producer i for different values of p and m.

Gross Expected Profit	$p = \frac{1}{6}$	$p = \frac{1}{4}$	$p = \frac{1}{2}$	$p = \frac{3}{5}$	$p = \frac{2}{3}$
$E\pi_i(g(M_i^0))$	516.66	475	350	300	266.66
$E\pi_i(g(M_i^a))$	913.88	1011.25	1065	986.4	902.21
$E\pi_i(g(M_i^b))$	1133.78	1278.43	1302.5	1168.8	1042.95

[21] See footnote 20.

$E\pi_i(g(M_i^c))$	1242.95	1411.32	1421.25	1260.18	1113.56
$E\pi_i(g(M_i^d))$	1329.75	1517.3	1503.12	1315.17	1147.92
$E\pi_i(g(M_i^e))$	1342.02	1541.13	1535.3	1338.88	1167.8

Table -3: Expected marginal profits of producer i for different values of p and m.

Expected Marginal Profit	$p=\frac{1}{6}$	$p=\frac{1}{4}$	$p=\frac{1}{2}$	$p=\frac{3}{5}$	$p=\frac{2}{3}$
$E\pi_i(g(M_i^a))-E\pi_i(g(M_i^0))$	397.22	536.25	715	686.4	635.55
$E\pi_i(g(M_i^b))-E\pi_i(g(M_i^a))$	219.9	267.18	237.5	182.4	140.74
$E\pi_i(g(M_i^c))-E\pi_i(g(M_i^b))$	109.18	132.89	118.75	91.38	70.61
$E\pi_i(g(M_i^d))-E\pi_i(g(M_i^c))$	87.11	105.98	81.86	54.99	37.36
$E\pi_i(g(M_i^e))-E\pi_i(g(M_i^d))$	11.95	24.13	32.2	23.71	16.87

What Table 3 shows is that in Example 3, $E\pi_i(g(M_i))$ is concave in m, given the value of p, and this is true for every value of p in this example. This implies that we can compute the maximal size of a path connected sub network, i.e., of a component, given the positive cost of forming a link, $c>0$, and probability of getting a low demand shock in the output markets, $p>0$. However, this concavity may not be generic.

Formally, we can comment on the curvature of $E\pi_i(g(M))$ if we know the sign of $\{E\pi_i(g(M'))-E\pi_i(g(M))\}-\{E\pi_i(g(M))-E\pi_i(g(M''))\}$, where $|M'|=m+1; |M|=m$; and $|M''|=m-1$ for all $m>1$. In general, we can find conditions under which $E\pi_i(g(M))$ is concave. Proposition 4 states these.

Proposition 4. i) Let m be even. Suppose that $2^{m-1}c_k(1-p)A_k >^m c_k(1-p)^2 A_k' +^{m-2}c_k A_k''$, and $^m c_{\frac{m}{2}}(1-p)A_{\frac{m}{2}}' <^{m-1} c_{\frac{m}{2}} \frac{1}{m-1} A$, for all $k = 1,2,3,....,\frac{m}{2}-1$. Then $E\pi_i(g(M))$ is concave in $|M|$.

ii) Let m be odd. Suppose that $2^{m-1}c_k(1-p)A_k >^m c_k(1-p)^2 A_k' +^{m-2}c_k A_k''$, and $2^{m-1}c_{k'}A_{k'} >^m c_{k'}(1-p)A_{k'}' +^{m-1}c_{k'+1}\frac{1}{m}pA$, for all $k = 1,2,3,....,\frac{m-1}{2}-1$ and $k' = \frac{m-1}{2}$. Then $E\pi_i(g(M))$ is concave in $|M|$.

Proof: See appendix.

4.4. Network Architecture.

We are interested in the architecture of non-empty components. The next proposition shows when the empty network is strategically stable.

Proposition 5. Suppose that $c > 0$. Let $(M, g(M))$ be an empty sub network of $(N, g(N))$. It is pair wise stable if and only if $E\pi_i(g(M)+ij) - E\pi_i(g(M)) < c$, $\forall i \in M$.

Proof: See appendix.

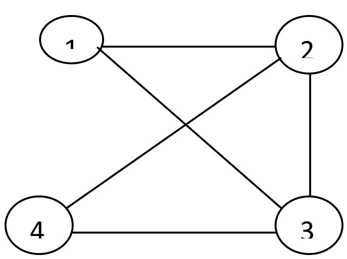
Proposition 5 says that if c is sufficiently high then the network $(N, g(N))$ will be empty, i.e., players will not form links with each other. The next proposition identifies a property of the architecture of trade credit guarantee networks when the cost of forming links, c, is positive, but less than $E\pi_i(g(M)+ij) - E\pi_i(g(M))$.

Proposition 6. Let $(M, g(M))$ be a component of the network $(N, g(N))$. Then $(M, g(M))$ has no cycles.

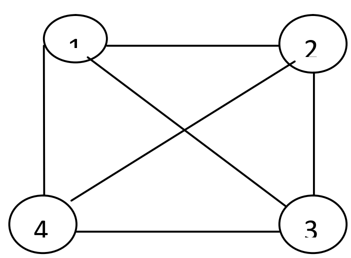
Proof: See appendix.

What Proposition 6 says is that no producer will form a link with another one if they are already indirectly path connected, irrespective of the length of the path. This is in sharp contrast with the result for distance based utility models where such links may be formed if the extra benefit from a direct link over the benefit from an indirect link is greater than the cost of forming a link.[22]

Proposition 6 has some important implications for the architecture of pair wise stable components, when c is positive. It rules out formation of complete networks, where all producers are linked to each other (Figure 6). It also rules out emergence of circular networks (Figure 1), triads (Figure 1), and cliques (Figure 6), since formation of these networks increases the costs to individual producers without increasing the benefit from being members of the network. What Proposition 6 does not rule out is the formation of a tree network. (Figure: 7).



6 (i) Clique



6 (ii) Complete Network

Figure 6: Clique and Complete network.

[22] See Propositions 6.3.1 and 6.3.2 in Jackson (2008).

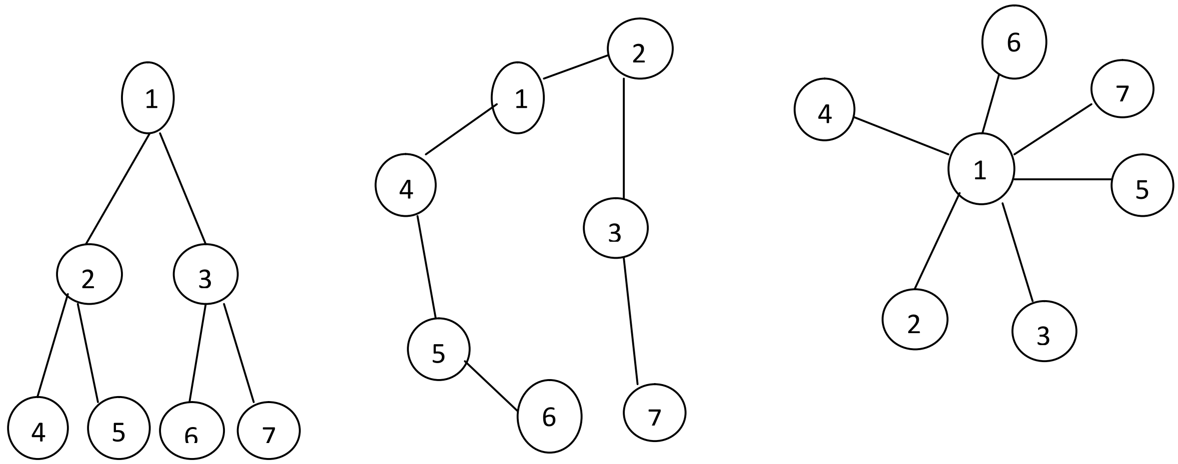


Figure: 7 (i) Tree Figure: 7 (ii) Line Figure: 7 (iii) Star

Figure 7. Three alternative network architectures of a tree network when m = 7

Since Proposition 4 tells us that the expected marginal benefit is a decreasing function of the size of a component, we are interested in knowing whether the cost of forming a link has any role in determining the optimal size of an acyclic component in our model. The next proposition shows that this is indeed true.

Proposition 7. Suppose that $c > 0$ and $E\pi_i(g(M_i))$ is concave in m, $m < n$. If $\{E\pi_i(g(M_i + ij)) - E\pi_i(g(M_i))\} < c < \{E\pi_i(g(M_i)) - E\pi_i(g(M_i - ik))\}$, $\forall i \neq k \in M_i$, where $j \in N - M_i$ then $(M_i, g(M_i))$ is a component of size $m < n$.

Proof: See appendix.

Corollary 1. Suppose that for all path connected sub-networks $(M, g(M))$ of $(N, g(N))$ with $c < \{E\pi_i(g(M + ij)) - E\pi_i(g(M))\}$. Then any component of $(N, g(N))$ has exactly n players and is not necessarily the network $(N, g(N))$.

Proof: See appendix.

An implication of Proposition 7 is that several components may co-exist in a network. The next proposition captures this intuitive notion and shows how many components can co-exist given the parameter values. Notice further, that while Proposition 6 rules out several architectural forms for these components, it does not rule out the tree architecture. Hence, while several components may co-exist in the network $(N, g(N))$, their architectures could be very different. That is, some components may be line networks, while others could be star networks. (See Figure 6.)

Proposition 8. Let α be the largest positive integer such that $\alpha m \leq n$. If (i) $\alpha \geq 1$ and $\alpha m = n$ then the network $(N, g(N))$ has exactly α components of size m, (ii) $\alpha \geq 1$ and $\alpha m < n$ then $(N, g(N))$ has α components of size m, and one component of size $n - \alpha m$.

Proof: Trivial□

What Corollary 1 and Proposition 8 say is that there is at most one component in the network $(N, g(N))$ whose size is not m. Further, the monotonicity of the marginal expected benefit, before deducting link formation costs, function ensures that m is a declining step function of c and, hence, the number of components in the network $(N, g(N))$ is an increasing step function of c.

So far we have confined our attention to the case where a member of a non-empty network can form links only with isolated players. In principle, forming a bridge between components generates more benefits than forming a link with an isolated player, since, a bridge links a player to a larger set of players than an isolated player. The next proposition deals with this issue.

Proposition 9. Let $(M_1, g(M_1))$ be a component and $(M_2, g(M_2))$ be a path connected sub-network of the network $(N, g(N))$ such that $|M_1| = m$ and $|M_2| = m' > 1$, and $M \cap M' = \phi$. Let $c > 0$. Then the path connected sub-network $(M_1 \cup M_2, (g(M_1 \cup M_2))$ is not pair wise stable.

Proof: See appendix.

Finally we investigate the efficiency property of these pair wise stable networks. In the literature, one sees a tension between pair wise stability and efficiency: a stable network need not be efficient, and vice versa[23]**.** Proposition 10 and Corollary 2 show that in our model this tension does not exist for path connected pair wise stable networks.

Proposition 10. Suppose that $c > 0$. Then pair wise stable, path connected sub-networks are efficient.

Proof: See appendix.

Proposition 10 says that the components of the network $(N, g(N))$ are individually efficient. Proposition 9 says that if the maximal size of a component of the network $(N, g(N))$ is $m < n$, then joining any two components leads to the formation of a sub-network that is not pair wise stable**.** This feature leads to Corollary 2.

Corollary 2. Suppose that $c > 0$. Then pair wise stable networks are efficient.

Proof: Trivial.

[23] See Jackson (2008) for an extensive introduction to this problem.

5. Concluding remarks.

In this paper we have assumed that enforcement problems do not arise. By itself, this reduces the role of a guarantor to that of an introducer only. However, in the absence of formal legal contracting, the guarantor, in effect, promises to compensate fully in the event of a default. The absence of enforcement problems, therefore, means that there is a mechanism in place that ensures that refusal to act as a guarantor (even an indirect one) and strategic default can be ruled out.

A second important feature of our model is that the size of the guaranteed amount does not decline with the number of links between the ultimate guarantor and the one taking the loan. This means that as long as the two players are linked through common friends or even a sequence of friends of friends the size of the amount guaranteed is the same as it would have been if they were directly linked. While this may appear to be a strong requirement, as Example 1 suggests, the results may not differ significantly if we allow for "small" decays. We have not modeled these aspects in this paper, but plan to do so in a subsequent paper.

Finally, we have gone against the abundant evidence that kinship ties and friendships outside the purely the professional do matter in the formation of economic networks by restricting ourselves to purely professional, perhaps professional acquaintance based ties. Even so, we find disjoint self-contained sub-networks, where all players are connected to each other can directly or otherwise, can co-exist. Some of these may have a central player (as in a star network), others may not. Further, as in Genicot and Ray (2003) there is a uniform upper bound on the size of the sub networks for purely strategic reasons. This means that even if strong kinship, or long term friendships between players exist, some kinsmen or friends may be left out of a sub network for purely strategic reasons, suggesting that strategic ties of the sort described here (and in Genicot and Ray (2003)) are supersede kinship and external friendship ties in the formation of such networks.

Appendix.

The full expression of equation $(5')$ is:

$$E\pi_i(g(M)) = {}^{m-1}c_1 p(1-p)^{m-1} A_1 + {}^{m-1}c_2 p^2(1-p)^{m-2} A_2 + {}^{m-1}c_3 p^3(1-p)^{m-3} A_3 + \dots\dots\dots\dots +$$
$${}^{m-1}c_k p^k(1-p)^{m-k} A_k + {}^{m-1}c_{\frac{m}{2}-1} p^{\frac{m}{2}-1}(1-p)^{\frac{m}{2}+1} A_{\frac{m}{2}-1} + \frac{1}{m-1}\,{}^{m-1}c_{\frac{m}{2}} p^{\frac{m}{2}}(1-p)^{\frac{m}{2}} A_{\frac{m}{2}} +$$
$$[(\frac{m}{2}-2)p^{\frac{m}{2}-1}(1-p)^{\frac{m}{2}-1} + (\frac{m}{2}-3)p^{\frac{m}{2}-2}(1-p)^{\frac{m}{2}-2} + \dots + (m-1-k)p^{m-k}(1-p)^{m-k} + \dots\dots +$$
$$p(1-p)][\pi(F)-\pi(0)] + (1-p)\pi(0) + p\pi_l$$

And the full expression of equation $(5'')$ is:

$$E\pi_i(g(M)) = {}^{m-1}c_1 p(1-p)^{m-1} A_1 + {}^{m-1}c_2 p^2(1-p)^{m-2} A_2 + {}^{m-1}c_3 p^3(1-p)^{m-3} A_3 + \dots\dots\dots\dots +$$
$${}^{m-1}c_k p^k(1-p)^{m-k} A_k + \dots\dots + {}^{m-1}c_{\frac{m-1}{2}} p^{\frac{m-1}{2}}(1-p)^{\frac{m-1}{2}+1} A_{\frac{m-1}{2}} + [(\frac{m-1}{2}-2)p^{\frac{m-1}{2}-1}(1-p)^{\frac{m-1}{2}-1} +$$
$$(\frac{m-1}{2}-3)p^{\frac{m-1}{2}-2}(1-p)^{\frac{m-1}{2}-2} + \dots + (m-1-k)p^{m-k}(1-p)^{m-k} + \dots\dots + p(1-p)][\pi(F)-\pi(0)] +$$
$$(1-p)\pi(0) + p\pi_l$$

where $A_1 = \pi(\min(1,\frac{1}{m-1})F) - \min(1,\frac{1}{m-1})\pi(F) - \{1-\min(1,\frac{1}{m-1})\}\pi(0);$

$$A_2 = \pi(\min(1,\frac{2}{m-2})F) - \min(1,\frac{2}{m-2})\pi(F) - \{1-\min(1,\frac{2}{m-2})\}\pi(0);$$
$$A_3 = \pi(\min(1,\frac{3}{m-3})F) - \min(1,\frac{3}{m-3})\pi(F) - \{1-\min(1,\frac{3}{m-3})\}\pi(0);$$

....

$$A_{\frac{m}{2}-1} = \pi(\min(1,\frac{m-2}{m+2})F) - \min(1,\frac{m-2}{m+2})\pi(F) - \{1-\min(1,\frac{m-2}{m+2})\}\pi(0);\ A = \pi(F)-\pi(0)\ .$$

Proof of Proposition 1. Suppose that m is even. In equation $(5')$ the term $p\pi_l + (1-p)\pi(0) > 0$ and is independent of the sub network size. Moreover, $\pi(F)-\pi(0) > 0$ by construction as F

□

maximizes $\pi(F)$ on $[0,\propto)$. The right hand size of the equation ($5'$) will be positive if the values of $A_k, \forall k = 1,2,...,\frac{m}{2}-1$ are positive. Now as $\pi_i(L_{cg})$ is concave in L_{cg}, all these terms are positive.

Using equation ($5''$) and noting that $k = 1,2,...,\frac{m-1}{2}$, an analogous argument establishes the theorem for odd values of m. □

The full expression of equation (6) is:

$$
\begin{aligned}
&E\pi_i(g(M'))={}^m c_1 p(1-p)^m[\pi\{\min(1,\frac{1}{m})F\}-\min(1,\frac{1}{m})\pi(F)-\{1-\min(1,\frac{1}{m})\}\pi(0)]\\
&+{}^m c_2 p^2(1-p)^{m-1}[\pi\{\min(1,\frac{2}{m-1})F\}-\min(1,\frac{2}{m-1})\pi(F)-\{1-\min(1,\frac{2}{m-1})\}\pi(0)]+\\
&{}^m c_3 p^3(1-p)^{m-2}[\pi\{\min(1,\frac{3}{m-2})F\}-\min(1,\frac{3}{m-2})\pi(F)-\{1-\min(1,\frac{3}{m-2})\}\pi(0)]+.........+\\
&{}^m c_k p^k(1-p)^{m-k}[\pi\{\min(1,\frac{k}{m-k})F\}-\min(1,\frac{k}{m-k})\pi(F)-\{1-\min(1,\frac{k}{m-k})\}\pi(0)]+........+\\
&{}^m c_{\frac{m}{2}-1} p^{\frac{m}{2}-1}(1-p)^{\frac{m}{2}+2}[\pi\{\min(1,\frac{m-2}{m+4})F\}-\min(1,\frac{m-2}{m+4})\pi(F)-\{1-\min(1,\frac{m-2}{m+4})\}\pi(0)]+\\
&{}^m c_{\frac{m}{2}} p^{\frac{m}{2}}(1-p)^{\frac{m}{2}+1}[\pi\{\min(1,\frac{m}{m+2})F\}-\min(1,\frac{m}{m+2})\pi(F)-\{1-\min(1,\frac{m}{m+2})\}\pi(0)]+\\
&+(\frac{m}{2}-1)p^{\frac{m}{2}}(1-p)^{\frac{m}{2}}[\pi(F)-\pi(0)]+(\frac{m}{2}-2)p^{\frac{m}{2}-1}(1-p)^{\frac{m}{2}-1}[\pi(F)-\pi(0)]+\\
&(\frac{m}{2}-3)p^{\frac{m}{2}-2}(1-p)^{\frac{m}{2}-2}[\pi(F)-\pi(0)]+(\frac{m}{2}-4)p^{\frac{m}{2}-3}(1-p)^{\frac{m}{2}-3}[\pi(F)-\pi(0)]+......+\\
&(\frac{m}{2}-\{\frac{m}{2}-1\})p^{(\frac{m}{2}-\{\frac{m}{2}-1\})}(1-p)^{(\frac{m}{2}-\{\frac{m}{2}-1\})}[\pi(F)-\pi(0)]+(1-p)\pi(0)+p\pi_l
\end{aligned}
$$

The full expression of equation (7)is:

$$E\pi_i(g(M')) - E\pi_i(g(M)) = mp(1-p)^m A_1' - (m-1)p(1-p)^{m-1} A_1 + {}^m c_2 p^2 (1-p)^{m-1} A_2' -$$
$${}^{m-1}c_2 p^2 (1-p)^{m-2} A_2 + {}^m c_3 p^3 (1-p)^{m-2} A_3' - {}^{m-1}c_3 p^3 (1-p)^{m-3} A_3 + \dots\dots\dots\dots + $$
$${}^m c_k p^k (1-p)^{m+1-k} A_k' - {}^{m-1}c_k p^k (1-p)^{m-k} A_k + {}^m c_{\frac{m}{2}-2} p^{\frac{m}{2}-2} (1-p)^{\frac{m}{2}+2} A_{\frac{m}{2}-2}' -$$
$${}^{m-1}c_{\frac{m}{2}-2} p^{\frac{m}{2}-2} (1-p)^{\frac{m}{2}+2} A_{\frac{m}{2}-2} + {}^m c_{\frac{m}{2}-1} p^{\frac{m}{2}-1} (1-p)^{\frac{m}{2}+2} A_{\frac{m}{2}-1}' - {}^{m-1}c_{\frac{m}{2}-1} p^{\frac{m}{2}-1} (1-p)^{\frac{m}{2}+1} A_{\frac{m}{2}-1} +$$
$${}^m c_{\frac{m}{2}} p^{\frac{m}{2}} (1-p)^{\frac{m}{2}+1} A_{\frac{m}{2}}'$$

where $A_1' = \pi(\min(1,\frac{1}{m})F) - \min(1,\frac{1}{m})\pi(F) - \{1-\min(1,\frac{1}{m})\}\pi(0);$

$A_1 = \pi(\min(1,\frac{1}{m-1})F) - \min(1,\frac{1}{m-1})\pi(F) - \{1-\min(1,\frac{1}{m-1})\}\pi(0);$

$A_2' = \pi(\min(1,\frac{2}{m-1})F) - \min(1,\frac{2}{m-1})\pi(F) - \{1-\min(1,\frac{2}{m-1})\}\pi(0);$

$A_2 = \pi(\min(1,\frac{2}{m-2})F) - \min(1,\frac{2}{m-2})\pi(F) - \{1-\min(1,\frac{2}{m-2})\}\pi(0);$

....

$A_{\frac{m}{2}-1}' = \pi(\min(1,\frac{m-2}{m+4})F) - \min(1,\frac{m-2}{m+4})\pi(F) - \{1-\min(1,\frac{m-2}{m+4})\}\pi(0)$

$A_{\frac{m}{2}-1} = \pi(\min(1,\frac{m-2}{m+2})F) - \min(1,\frac{m-2}{m+2})\pi(F) - \{1-\min(1,\frac{m-2}{m+2})\}\pi(0)$

$A_{\frac{m}{2}}' = \pi(\min(1,\frac{m}{m+2})F) - \min(1,\frac{m}{m+2})\pi(F) - \{1-\min(1,\frac{m}{m+2})\}\pi(0);$ $A_{\frac{m}{2}} = \pi(F) - \pi(0)$.

Further simplifying equation (7) we get the following:

$$E\pi_i(g(M')) - E\pi_i(g(M)) = mp(1-p)^m A_1' + (m-1)p(1-p)^{m-1}(\frac{m}{2}pA_2' - A_1) +$$

$$p^2(1-p)^{m-2}\frac{(m-1)(m-2)}{2}(\frac{m}{3}pA_3' - A_2) + p^3(1-p)^{m-3}\frac{(m-1)(m-2)(m-3)}{3.2.1}(\frac{m}{4}pA_4' - A_3) +$$

$$.................. + p^{\frac{m}{2}-2}(1-p)^{\frac{m}{2}+2}\frac{(m-1)(m-2).........(\frac{m}{2}+2)}{(\frac{m}{2}-2)(\frac{m}{2}-3)...3.2.1}(\frac{m}{\frac{m}{2}-1}pA_{\frac{m}{2}-1}' - A_{\frac{m}{2}-2}) +$$

$$p^{\frac{m}{2}-1}(1-p)^{\frac{m}{2}+1}\frac{(m-1)(m-2.........(\frac{m}{2}+1)}{(\frac{m}{2}-1)(\frac{m}{2}-2)...3.2.1}(\frac{m}{\frac{m}{2}}pA_{\frac{m}{2}}' - A_{\frac{m}{2}-1})$$

Proof of Proposition 2. Suppose that m is an even number. Notice that the first term of right hand side of Equation (7) is positive as $\pi(L_{cg})$ is concave in L_{cg} and $A_k' > A_{k-1}$ for all $k = 2,3,4,..........\frac{m}{2}$. The sign of right hand side is positive if the sign of the term within the first bracket, $(\frac{m}{k}pA_k' - A_{k-1})$, is positive. We denote the value of p for which $(\frac{m}{k}pA_k' - A_{k-1})$ is positive by $p_k \geq (\frac{k}{m})(\frac{A_{k-1}}{A_k'})$. Therefore, if $p > \max(p_k)$ then $E\pi_i(g(M_i')) - E\pi_i(g(M_i)) > 0$. □

Using equations $(5'')$ and $(6')$, and noting that $k = 2,3,...,\frac{m-1}{2}$ an analogous argument establishes the proposition for odd values of m. □

Proof of Proposition 3. Notice Proposition 2 shows that there is size monotonicity in this model and given the Assumption 2 the gross expected profit depends on the number of producers in a sub network only. Therefore as long as two producers are path connected in a sub network, it does not matter who among them forms a new link. Hence, $E\pi_i(g(M_i) + sj) > E\pi_i(g(M_i))$. □

Proof of Proposition 4. **i)** Let m be even**.** $E\pi_i(g(M))$ is concave if $\{E\pi_i(g(M'))-E\pi_i(g(M))\}-\{E\pi_i(g(M))-E\pi_i(g(M''))\}<0$. Now

$$\{E\pi_i(g(M'))-E\pi_i(g(M))\}-\{E\pi_i(g(M))-E\pi_i(g(M''))\}$$

$$=\sum_{k=1}^{\frac{m}{2}-1} p^k(1-p)^{m-1-k}[{}^m c_k(1-p)^2 A_k{}' + {}^{m-2}c_k A_k{}'' - 2^{m-1}c_k(1-p)A_k] +$$

$$p^{\frac{m}{2}}(1-p)^{\frac{m}{2}-1}[{}^m c_{k'}(1-p)A_{\frac{m}{2}}{}' - {}^{m-1}c_{\frac{m}{2}}\frac{1}{m-1}A]$$

where, $A_k{}'' = \pi(\min(1,\frac{k}{m-1-k})F)-\min(1,\frac{k}{m-1-k})\pi(F)-\{1-\min(1,\frac{k}{m-1-k})\}\pi(0)$

Inspection of this equation shows that the theorem is true when $2^{m-1}c_k(1-p)A_k > {}^m c_k(1-p)^2 A_k{}' + {}^{m-2}c_k A_k{}''$, and ${}^m c_{\frac{m}{2}}(1-p)A_{\frac{m}{2}}{}' < {}^{m-1}c_{\frac{m}{2}}\frac{1}{m-1}A$, for all values of $k=1,2,3,....,\frac{m}{2}-1$.

ii) Let m be odd. $E\pi_i(g(M))$ is concave if $\{E\pi_i(g(M'))-E\pi_i(g(M))\}-\{E\pi_i(g(M))-E\pi_i(g(M''))\}<0$. Now

$$\{E\pi_i(g(M'))-E\pi_i(g(M))\}-\{E\pi_i(g(M))-E\pi_i(g(M''))\}$$

$$=\sum_{k=1}^{\frac{m-1}{2}-1} p^k(1-p)^{m-1-k}[{}^m c_k(1-p)^2 A_k' + {}^{m-2}c_k A_k{}'' - 2^{m-1}c_k(1-p)A_k] +$$

$$p^{\frac{m-1}{2}}(1-p)^{\frac{m-1}{2}-1}[{}^m c_{\frac{m-1}{2}}(1-p)A_{\frac{m-1}{2}}{}' + \frac{1}{m}{}^m c_{\frac{m-1}{2}+1}pA - 2^{m-1}c_{\frac{m-1}{2}}A_{\frac{m-1}{2}}]$$

Inspection of this equation shows that the theorem is true when $2^{m-1}c_k(1-p)A_k{}' > {}^m c_k(1-p)^2 A_k + {}^{m-2}c_k A_k{}''$, and $2^{m-1}c_{k'}A_{k'} > {}^m c_{k'}(1-p)A_{k'}{}' + {}^m c_{k'+1}\frac{1}{m}pA$, where $k' = \frac{m-1}{2}$, and $k = 1,2,3,....,\frac{m-1}{2}-1$. □

Proof of Proposition 5. Notice first that since gross expected profit functions are the same for all producers $i \in M$, $\{E\pi_i(g(M)+ij) - E\pi_i(g(M))\} = \{E\pi_j(g(M)+ji) - E\pi_j(g(M))\}$, $\forall i \neq j \in M$. Suppose that $E\pi_i(g(M)+ij) - E\pi_i(g(M)) < c$. Then neither producer i nor producer j has an incentive to form the link between themselves. To establish the only if part, suppose that $(M, g(M))$ is pair wise stable. This means that for any pair ij in M, $E\pi_i(g(M)+ij) - E\pi_i(g(M)) < c$. Further since $(M, g(M))$ is an empty sub network, there is no link to delete. □

Proof of Proposition 6. Suppose $(M, g(M))$ has a cycle and producers i and j, who are indirectly path connected in the component $(M, g(M))$, have a direct link between them as well. Since $E\pi_i(g(M)-ij) = E\pi_i(g(M))$, producers i and j can save the cost $c > 0$ without any loss in benefit by deleting the direct link between them and maintain the indirect link between them as the benefit does not decay with path length between them (A3). Therefore, given the positive cost of forming a link, no two producers i and j, who are indirectly path connected in the component, have any incentive to maintain this direct link between them.

□

Proof of Proposition 7. The condition $\{E\pi_i(g(M_i + ij)) - E\pi_i(g(M_i))\} < c <$ $\{E\pi_i(g(M_i)) - E\pi_i(g(M_i - ik))\}$, where $j \in N - M_i$ and $i \neq k \in M_i$, guarantees that the path connected sub network $(M_i, g(M_i))$ is stable. Concavity of $E\pi_i(g(M_i))$ guarantees that the size of such a stable sub- network is unique. □

Proof of Corollary 1. The condition $c < \{E\pi_i(g(M + ij)) - E\pi_i(g(M))\}$ implies that there is no path connected sub network $(M, g(M))$ of $(N, g(N))$ that is pair wise stable, as addition of one more player in the sub network is profitable for each player in M. Addition of players in a sub network stops when the number of players in it reaches n. Since the resulting sub network is stable, then by Proposition 6 it has no cycles. Since, the network $(N, g(N))$ may have cycles, a component of $(N, g(N))$ need not be equal to $(N, g(N))$. □

Proof of Proposition 9. Let $i \in M_1$ and $j \in M_2$. The link ij is profitable for both players if and only if $\{E\pi_i(g(M_1 \cup M_2)) - E\pi_i(g(M_1)\} > c$ and $\{E\pi_j(g(M_2 \cup M_1)) - E\pi_j(g(M_2)\} > c$. Suppose this is true. The resulting path connected sub network $(M_1 \cup M_2, g(M_1 \cup M_2))$ has $m + m' > m$ players. Since the sub network $(M_1, g(M_1))$ is a component of the network $(N, g(N))$ by Proposition 7, its size m is the maximum size of a stable path connected sub network. Hence, by Proposition 7, the path connected sub network $(M_1 \cup M_2, g(M_1 \cup M_2))$ can not be stable. □

Proof of Proposition 10. Let $(M, g(M))$ be any component of the network $(N, g(N))$. From Proposition 6, this is a tree with k players, where from Propositions 7, 8 and Corollary 1, $k \in \{m, n-\alpha m, n\}$. Any alternative path connected non-empty network on the set of M has at least as many links and all of them generate the same benefit. Hence, the total cost of link formation for the component $(M, g(M))$ is no greater than that on any other path connected network on M, and the total benefits remain the same. □

References.

1) Atanasova, C. (2007), “Access to Institutional Finance and the Use of Trade Credit”, Financial Management, Vol. 36, No. 1, pp. 49-67.
2) Bala, V. and S. Goyal. (2000), “A Non cooperative Model of Network Formation”, Econometrica, Vol. 68, No. 5, pp. 1181-1229.
3) Belleflamme, P. and F. Bloch. (2004), “Market Sharing Agreements and Collusive Networks”, International Economic Review, Vol. 45, No. 2, pp. 387-411.
4) Bendor, J. and D. Mookherjee. (1990), “Norms, Third - Party Sanctions, and Cooperation”, Journal of Law, Economics, and Organization, Vol. 6, No. 1, pp. 33-63.
5) Bian, Y. (1997), “Bringing Strong Ties Back in: Indirect Ties, Network Bridges, and Job Searches in China”, American Sociological Review, Vol.62, No. 3, pp. 366-385.
6) Bloch, F. and B. Dutta. (2009), “Communication Networks with Endogeneous Link Strength”, Games and Economic Behaviour, Vol. 66, pp. 39-56.
7) Bloch, F., G. Genicot, and D. Ray. (2008), “Informal insurance in social networks”, Journal of Economic Theory, Vol. 143, pp. 36-58.
8) Boorman, S. A. (1975), “A Combinatorial Optimization Model for Transmission of Job Information Through Contact Networks”, The Bell Journal of Economics, Vol. 6, No. 1, pp. 216-249.
9) Bramoulle´, Y. and R. Kranton. (2007), “Risk-sharing networks”, Journal of Economic Behavior & Organization, Vol. 64, No. , pp. 275–294.
10) Burkart, M. and T. Ellingsen. (2004), “In-kind Finance: A Theory of Trade Credit”, The American Economic Review, Vol. 94, No. 3, pp. 569-590.
11) Calvo-Armengol, A. (2004), “Jon contact networks”, Journal of Economic Theory, Vol.5, No. 1, pp. 191-206.

12) Calvo-Armengol, A.C. and Y. Zenou. (2004), "Job Matching, Social Network, and Word-of-Mouth Communication", Journal of Urban Economics, Vol. 57, pp. 500-522.

13) Giannetti, M., M. Burkart. and T. Ellingsen. (2011), "What you sell is what you lend? Explaining Trade Credit Contracts", The Review of Financial Studies, Vol. 24, No. 4, pp. 1261-1298.

14) Coate, S. and M. Ravallion. (1993), "Reciprocity without commitment: Characterization and performance of informal insurance arrangements", Journal of development Economics, Vol. 40, No. , pp. 1-24.

15) Chatterjee, K. and B. Dutta. (2011), "Credibility and Strategic Learning in Networks", mimeo.

16) Cheng, N. S. and R. Pike. (2003), "The Trade Credit Decision: Evidence of UK firms", Managerial and Decision Economics, Vol. 24, No. 6/7, pp.419-438.

17) Cull, R. *et al* (2009), "Formal finance and trade credit during China's transition", Journal of Financial Intermediation, Vol. 18, pp. 173-192.

18) De Weerdt, J. and S. Dercon, (2006), "Risk-sharing networks and insurance against illness", Journal of Development Economics, Vol. 81, pp. 337–356.

19) Diestel, R. (2005), Graph Theory, Third Edition, Springer-Verlag Heidelberg, New York.

20) Dutta, B. and S. Muthuswami. (1997), "Stable Networks", Journal of Economic Theory, Vol. 76, No. , pp. 322-344.

21) Ellison, G. and D. Fudenberg. (1995), "Word- of- Mouth Communication and Social Learning", The Quarterly Journal of Economics, Vol. 110, No.1, pp. 93-125.

22) Fafchamps, M. (1997), "Trade Credit in Zimbabwean manufacturing", World Development, Vol. 25, No. 5, pp. 795-815.

23) Fafchamps, M. (2000), “Ethnicity and credit in African manufacturing”, Journal of Development Economics, Vol. 61, No. , pp. 205-235.

24) Fafchamps, F. and S. Lund. (2003), “Risk-sharing networks in rural Philippines”, Journal of Development Economics, Vol. 71, pp. 261-287.

25) Fafchamps, F. and F. Gubert, (2007), “The formation of risk sharing networks”, Journal of Development Economics, Vol. 83, pp. 326–350.

26) Fisman, R. and I. Love. (2003), “Trade Credit, Financial Intermediary Development, and Industry Growth, The Journal of Finance, Vol. 58, No. 1, pp. 353-374.

27) Fisman, R. and M. Raturi. (2004), “Does Competition Encourage Credit Provision? Evidence from African Trade Credit Relationships”, The Review of Economics and Statistics, Vol. 86, No. 1, pp. 345-352.

28) Galeotti, A. and S. Goyal. (2002), “Network Formation with Heterogeneous Players”, TI 2002-069/1, Tinbergen Institute Discussion Paper.

29) Galeotti, A., S. Goyal, and J. Kampherst. (2006), “Network Formation with Heterogeneous Players”, Games and Economic Behaviour , Vol. 54, No. 2, pp . 353-373.

30) Genicot, G. and D. Ray. (2003), “Group Formation in Risk-Sharing Arrangements”, The Review of Economic Studies, Vol. 70, No. 1, pp. 87-113.

31) Goyal, S. and S Joshi. (2003): “Networks of Collaboration in Oligopoly”, Games and Economic Behaviour, Vol. 43, pp. 57-85.

32) Goyal, S. and J. Gonzalez. (2001): “R & D Networks”, The Rand Journal of Economics, Vol. 32, No. 4, pp. 686-707.

33) Goyal, S. (2015), “Networks In Economics: A Perspective On The Literature”, Cambridge- INET Institute, Working Paper Series No. 2015/05.

34) Granovetter, M. S. (1973), “The Strength of Weak Ties”, American Journal of Sociology, Vol. 78, No.6, pp. 1360-1380.

35) Jackson, M.O. and A. Wolinsky (1996), “A Strategic Model of Social and Economic Networks”, Journal of Economic Theory, Vol. 71, pp. 44-74.

36) Jackson, M. O. and A. Watts (2002), “The Evolution of Social and Economic Networks”, Journal of Economic Theory, Vol. 106, pp. 265-295.

37) Jackson, M.O. (2003), The Stability and Efficiency of Economic and Social Networks,. in Advances in Economic Design, edited by S. Koray and M. Sertel, Springer-Verlag: Heidelberg, and reprinted in Networks and Groups: Models of Strategic Formation, edited by B. Dutta and M.O. Jackson, Springer- Verlag: Heidelberg.

38) Jackson, M. (2005), “Allocation rules for network games”, Games and Economic Behaviour, Vol. 51, No. 1, pp. 128-154.

39) Jackson, M. (2008), Social and Economic Networks, Princeton University Press.

40) Jackson, M., T. Barraquer and X. Tan. (2012), “Social Capital and Social Quilts: Network Patterns of Favor Exchange”, The American Economic Review, Vol. 102, No. 5, pp. 1857-1897.

41) Jackson, M. (2014), “Networks in the Understanding of Economic Behaviours”, The Journal of Economic Perspectives, Vol. 28, No. 4, pp. 3-22.

42) Kranton, R. and D. Minehart. (2000), “Networks versus vertical integration”, Rand Journal of Economics, Vol. 31, No. 3, pp. 570-601.

43) Kranton, R. and D. Minehart. (2001), “A Theory of Buyer-Seller Networks”, The American Economic Review, Vol. 91, No. 3, pp. 485-508.

44) McMillan, J. and C. Woodruff. (1999), "Interfirm Relationships and Informal Credit in Vietnam", The Quarterly Journal of Economics, Vol. 114, No. 4, pp. 1285-1230.

45) Montgomery, J. D. (1991), "Social Networks and Labour-Market Outcomes: Toward an Economic Analysis", The American Economic Review, Vol. 81, No. 5, pp. 1408-1418.

46) Montgomery, J. D. (1992), "Job search and network composition: Implications of the strength- of- weak- ties hypothesis", American Sociological Review, Vol. 57, pp. 586-596.

47) Murgai, R., P. Winters., E. Sadoulet., and A. de Janvry. (2002), "Localized and incomplete mutual insurance", Journal of Development Economics, Vol. 67, No. 2, pp. 245-274.

48) Petersen, M. A. and R.G. Ragan. (1997), "Trade Credit: Theories and Evidence", The Review of Financial Studies, Vol. 10, No. 3, pp. 661-691.

49) Watts, A. (2001), "A Dynamic Model of Network Formation", Games and Economic Behaviour, Vol. 34, pp. 331-341.

50) Yakubovich, V. (2005), "Weak Ties, Information, and Influence: How Workers Find Jobs in a Local Russian Labor Market", American Sociological Review, Vol. 70, No.3, pp. 408-421.